AlphaLab-Risk: Machine Learning for Portfolio Risk Forecasting and Stress Resilience

Research report, draft v2 (2026-11-12) · supersedes draft v1 (2026-11-07)

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Code, data provenance and full history: `github.com/Guohuan2026/alphalab-risk` (private), branch `alphalab-risk`. Every number below was re-checked against the saved result files (`data/results/`) for this draft; the log of results in the order they were seen is `docs/RESULTS_LOG.md`.

> **How to read this report.** Each claim carries one of four evidence labels, defined in §4.6 and justified test by test in `docs/EVIDENCE_STATUS.md`: **[Inference]** a test whose finite-sample size was checked and found acceptable for that sample length and level; **[Descriptive]** a measured difference, rank or rate with no claim of statistical significance; **[Exploratory]** an analysis added after the held-out lockbox was opened (development data only); **[Unsupported test]** a test that failed its size check, whose statistic is reported only as a description. Draft v1 used Diebold–Mariano, Model Confidence Set and HAC-regression p-values as evidence of significance; a frozen calibration study (§6.1) showed these tests over-reject badly on data like ours, so every such statement has been withdrawn or restated.

Abstract

Can machine-learning (ML) models forecast portfolio downside risk better than the econometric methods used by banks and asset managers, particularly during market stress? I compare eleven one-day Value-at-Risk (VaR) and Expected Shortfall (ES) models on US equity (SPY, the headline portfolio), a 60/40 multi-asset portfolio and Canadian equity (XIU.TO), using strictly out-of-sample walk-forward forecasts, a design frozen before any GARCH, ML or lockbox result existed (only baseline historical-simulation and EWMA results had been seen), and a 2023–2026 lockbox opened exactly once. In the 1997–2022 development sample, the ML models' VaR forecasts were rejected by Christoffersen's conditional-coverage test in 11 or 12 of 12 portfolio × level cells, while the conditional-coverage null for filtered historical simulation with GJR-GARCH volatility (FHS-GJR) was not rejected in 11 of 12 [Inference]. No ML model's average joint VaR–ES (FZ0) loss was lower than FHS-GJR's by more than 0.002 in any P1 development cell, and each ML model's loss relative to FHS-GJR was more adverse in crisis episodes than in calm periods in all 36 model × setting × level cells [Descriptive]. A mechanism is visible directly: tree models cannot forecast risk outside their training range, so in October 2008 the forest-based VaR stayed flat while VIX reached 70. In the calm lockbox the average-loss ordering reversed: gradient boosting had lower mean FZ0 than FHS-GJR in every portfolio and level, so the pre-stated replication rule returned **NOT REPLICATED** [Descriptive, by a pre-stated rule]. FHS-GJR's conditional-coverage null was again not rejected, but so were those of most other models, with only about ten expected breaches at 99%. Before interpreting the comparisons I checked every test's size by simulation under realistic persistence; the pairwise and multiple-comparison tests (Diebold–Mariano, Model Confidence Set) and the stress regression failed, so the study's comparative findings are descriptive. The evidence is consistent with a limited, regime-dependent role for ML (the hybrid that uses ML only for volatility had lower loss than FHS-GJR in calm periods; direct ML tail models did not), but it does not establish that any model is statistically superior.

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1. Research question and contribution

> Can machine-learning models improve the accuracy and robustness of portfolio downside-risk forecasts compared with traditional econometric methods, particularly during periods of financial market stress?

Why it matters. Banks report 99% VaR and 97.5% ES under the Basel FRTB framework, and pension funds and asset managers set risk budgets from the same quantities. ML methods are increasingly proposed as replacements, but published comparisons often lack strict out-of-sample discipline, proper scoring rules, or an examination of behaviour in crises, the regime where risk models matter most.

Hypotheses (pre-registered at git tag `prereg-v1`):

	Hypothesis	Pre-registered evidence	Status in this draft
H1	ML VaR forecasts are correctly calibrated out-of-sample.	Kupiec, Christoffersen	**Not supported for ML** [Inference]
H2a	ML has lower joint VaR–ES loss than the best econometric model.	Mean FZ0, DM, MCS	**No evidence for H2a**; differences descriptive (DM/MCS unsupported)
H2b	Any ML gain comes from non-linearity, not merely from extra inputs (vs a linear model on the same features).	Mean FZ0, DM vs the lasso control	**No evidence for H2b**; descriptively reversed in the deep tail
H3	Relative performance differs in stress periods.	Breach rates by regime; HAC stress-dummy regression	**Consistent with H3** [Descriptive]; not established by a validated test
H4	ML forecasts respond to economically meaningful drivers, stably.	Three pre-stated criteria	**Not met** under the all-three rule [Descriptive]

Contributions.

1. A like-for-like comparison: same data, horizon, confidence levels, information set and tuning protocol for every model.
2. A control model separating "better inputs" from "non-linearity" (H2b).
3. A stress analysis with ex-ante episodes, a VIX regime and reaction-speed event studies, and a concrete mechanism for why tree-based tail models fail in crises (saturation).
4. A frozen finite-sample calibration of every backtest used, which decided how strongly each result may be read (§6.1). This is a check of known asymptotic tests on realistic designs, not a new statistical method.
5. A research-integrity process (design freeze, pre-committed clarifications, results log, once-only lockbox, bit-identical reproduction, versioned amendments) that a model-validation team can audit.

2. Background

VaR and ES. With loss $L = -r$ and tail probability α, VaR_α is the loss exceeded with probability α, and $ES_\alpha = E[L \mid L \geq VaR_\alpha]$ is the mean loss beyond it. ES captures tail severity, which VaR ignores; that is why FRTB moved capital to ES at 97.5%.

Elicitability. To *rank* forecasts we need a scoring rule whose expected value is minimised by the true forecast. Quantiles (VaR) have one: the tick loss. ES alone does not (Gneiting 2011), but the pair (VaR, ES) does (Fissler & Ziegel 2016). I use the FZ0 loss of Patton, Ziegel & Chen (2019); lower (more negative) is better.

Models. GARCH (Bollerslev 1986) and GJR-GARCH (Glosten, Jagannathan & Runkle 1993) capture volatility clustering and the leverage effect; FHS (Barone-Adesi et al. 1999) combines GARCH volatility with empirical residual tails; HAR (Corsi 2009) models volatility over daily, weekly and monthly horizons; quantile regression forests (Meinshausen 2006) and gradient boosting with the pinball loss are the main ML quantile learners.

Backtests and their known weak points. Coverage tests (Kupiec 1995; Christoffersen 1998) and the Dynamic Quantile test (Engle & Manganelli 2004) rely on χ^2 approximations that are known to be imprecise with few breaches; DQ in particular is over-sized at low α and moderate samples (Dumitrescu, Hurlin & Pham 2012). Diebold–Mariano (1995) tests and the Model Confidence Set (Hansen, Lunde & Nason 2011) rely on heteroskedasticity-and-autocorrelation consistent (HAC) or block-bootstrap approximations whose accuracy depends on how persistent the loss differentials are; fixed-b asymptotics (Kiefer & Vogelsang 2005) are one response. These known issues motivated the calibration study in §6.1.

3. Data

Portfolio	Composition	Calendar	Out-of-sample development	Lockbox
P1 (headline)	SPY total return (dividends included)	NYSE	targets 1997-01-15 → 2022-12-30 (6,535 forecasts)	2023-01-03 → 2026-10-30 (≈ 961)
P2	40% SPY, 20% EFA, 30% IEF, 10% LQD, monthly rebalance	NYSE	2006-07 → 2022-12 (4,142)	same
P3	XIU.TO (S&P/TSX 60), CAD	TSX	2003-09 → 2022-12 (4,838)	same

Sources. Prices, dividends and VIX from Yahoo Finance; credit (Moody's Baa − 10y) and term (10y − 3m) spreads from FRED. Every download is an immutable snapshot with a SHA-256 hash; each build records the snapshot it used.

Features (17, all stamped with information published by the end of day t): realized variance over 1/5/22/66 days (HAR structure), Parkinson and Garman–Klass range variance, drawdown from the forward-built wealth index, VIX and its 5-day change, and the two spreads with their 22-day changes. FRED series are lagged one session (they are published the next morning).

Data quality. Total returns are built from raw closes plus dividends, never from Yahoo's backward-adjusted levels, which embed future information. Three XIU.TO year-end distributions that were reinvested rather than paid are treated as non-cash (an explicit, logged correction; raw data untouched). SPY tracks the S&P 500 total-return index to within 0.13%/yr, but daily agreement is weaker before 2009 (a documented open limitation).

4. Methodology

4.1 The forecast contract

All models forecast the next day's loss at $\alpha \in \{1\%, 2.5\%, 5\%\}$ (99%, 97.5% and 95% VaR/ES). Main specification: expanding estimation window with at least 1,000 sessions, parameters re-estimated every 21 sessions, filters updated daily. Feature-based models refit and re-tune yearly (Deviation 3), which handicaps ML relative to the monthly econometric refits. A walk-forward engine produces one forecast per day using only data up to that day. An engine-level truncation test checks that a forecast computed on data cut off at date d equals the one computed in the full run.

4.2 Models

Family	Model	VaR / ES from
Baselines	HS-250, HS-500	empirical quantile and tail mean of past losses
	EWMA (RiskMetrics, $\lambda = 0.94$)	normal quantiles × EWMA volatility
Econometric	GARCH(1,1)-t, GJR-GARCH(1,1)-t	Student-t quantiles × GARCH volatility (`arch`)
	FHS-GJR	GJR volatility × empirical quantiles of standardised residuals
Linear	HAR-FHS	OLS volatility on 1/5/22-day realized variance + FHS tails
	Lasso QR (H2b control)	L1-penalised linear quantile regression on the 17 features
ML	Quantile regression forest (QRF)	forest leaf distributions (`quantile-forest`)
	Gradient boosting (GBQ)	one pinball-loss model per quantile level (scikit-learn)
	RF-volatility + FHS ("hybrid")	random forest for $\lvert r \rvert$ (a volatility proxy) × out-of-bag residual tails

ES for the quantile models is the average of predicted quantiles at $\alpha(k - ½)/5$, $k = 1…5$. A safety floor (VaR ≥ 1 bp, ES ≥ VaR) applies to every model (Deviation 2).

Tuning. One shared protocol for all ML models and the control. Hold out the last 20% of each training window with a one-day embargo, evaluate small grids fixed in advance with the mean tick loss, refit the best configuration on the whole window. Seeds are fixed.

4.3 Leakage safeguards

Every feature passes a truncation test, and the label (next-day loss) is checked to *fail* it. Scalers and clipping bounds are fitted on training data only. Stress regimes are used only to split the evaluation, never as inputs. The lockbox was enforced in code: forecasts into it required an explicit flag.

4.4 Evaluation

- **Calibration:** Kupiec (breach frequency, "POF"), Christoffersen independence ("IND") and conditional coverage ("CC"), Engle–Manganelli Dynamic Quantile ("DQ": are breaches predictable from past breaches or from the VaR level itself?), Basel traffic light.
- **ES:** McNeil–Frey exceedance-residual test (bootstrap) and its severity ratio (realised mean tail loss / forecast ES); Acerbi–Székely Z2 (frequency and severity of breaches together; Monte Carlo p-value under a shape-matched Student-t null, Clarification 7).
- **Ranking:** mean FZ0 and tick loss; Diebold–Mariano tests with Newey–West errors (lag $\lfloor 4(n/100)^{2/9} \rfloor$); the Model Confidence Set at 90%.
- **Stress:** eight ex-ante episodes (by target date) and a VIX-above-30 regime (at the forecast origin); loss differentials regressed on an episode dummy with HAC errors; an event study of VaR reaction after the largest losses.
- **Interpretability:** permutation importance on each yearly validation window; TreeSHAP attributions to VaR; three pre-stated criteria (concentration, economic sign, stability).

Before market data, every test was checked on simulated GJR-GARCH-t data with known true VaR and ES (FZ0 ranks the true model first; tests roughly correctly sized *under independence*). That check did not cover persistent, heavy-tailed loss differentials, which is why §6.1 was added.

4.5 Research integrity

The design was frozen at tag `prereg-v1` before any GARCH, ML or lockbox result existed (only baseline HS/EWMA results had been seen, and they are logged). Later decisions were committed before the results they govern (Deviations and Clarifications in `RESEARCH_DESIGN.md`; timeline in Appendix A). Every result is logged in the order it was seen. The lockbox protocol, including the rule for "replicated", was pushed to GitHub before the lockbox was opened. Everything added after the lockbox was opened (DQ, Z2, the ten-day table, the ES regression work and the calibration study) is labelled exploratory and uses development data or simulation only.

4.6 Evidence classification

Label	Meaning	Tests
[Inference]	Size within the frozen tolerance (≤ 6.4% at nominal 5%) in every gated simulation design for that sample length and α	Christoffersen CC (all cells); IND (except lockbox length at 5%); Kupiec (except lockbox length at 1%); DQ at 5% (and 2.5% at T ≈ 6,500)
[Descriptive]	Measured means, ranks, rates, coefficients; no significance claim	FZ0/tick means and ranks, hit rates, severity ratios, regime breach rates, SHAP, horizon table
[Exploratory]	Added after the lockbox was opened	DQ, Z2, ES regression, ten-day table, calibration study
[Unsupported test]	Failed its size check; statistic shown only as a description	Diebold–Mariano, MCS, HAC stress regression, Z2, McNeil–Frey (most cells), DQ at 1%, ES regression (retired)

"Supported" means size control within the simulated designs (§6.1); it says nothing about power. P2 and P3 (≈ 4,100–4,800 forecasts) lie between the two simulated lengths (961 and 6,535); a test that passed at both is treated as *provisionally* supported there, by endpoint checks: an interpolation heuristic, not guaranteed size control at those lengths. Kupiec's POF was gated only on designs with independent breaches; under clustered breaches it over-rejects even when the overall breach rate is right, so CC is the primary test. Statistics of unsupported tests are quoted as descriptions only; where a p-value from such a test is mentioned, it is to document what draft v1 claimed.

5. Results

5.1 Calibration (H1)

P1, development (6,535 forecasts; ≈ 65 expected breaches at 1%).

Model	1% hits	2.5% hits	5% hits	CC p (1 / 2.5 / 5%) [Inference]
FHS-GJR	**1.13%**	**2.86%**	**5.37%**	**0.32 / 0.15 / 0.36**
HAR-FHS	1.29%	2.98%	5.81%	<0.001 / 0.014 / 0.012
GJR-GARCH-t	1.39%	3.52%	6.12%	0.004 / <0.001 / <0.001
RF-vol + FHS	1.45%	3.05%	5.91%	<0.001 / 0.008 / <0.001
Lasso QR (control)	1.65%	3.55%	6.63%	<0.001 at every level
Quantile forest	1.76%	3.40%	6.06%	<0.001 at every level
Gradient boosting	2.92%	5.31%	8.28%	<0.001 at every level
EWMA	2.08%	3.64%	5.95%	<0.001 / <0.001 / 0.002
HS-250	1.56%	3.43%	5.71%	<0.001 at every level

Across the four development settings (P1, P2, P3, P1 rolling window; 12 cells per model), CC rejections at 5% [Inference]: FHS-GJR 1 of 12 (P2 at 1%: hit rate 1.40%, $p = 0.027$); HAR-FHS 6; lasso control 9; quantile forest, RF-vol hybrid and EWMA 11; gradient boosting, GARCH-t, GJR-t and both HS models 12. On P1, FHS-GJR is the only model not rejected by Kupiec or Christoffersen at any level; on P3, HAR-FHS and the lasso control are also never rejected by CC.

H1 is not supported for the ML models [Inference]: each is rejected in 11 or 12 of 12 cells. For FHS-GJR the correct reading is that *its conditional-coverage null was not rejected* in 11 of 12 development cells; that is a failure to find miscalibration with a test of known size, not proof of correct calibration.

Observed diagnostics and interpretation [Descriptive]. HS's breaches cluster (rejected by IND in all 12 cells), consistent with slow reaction. EWMA's realised tail losses exceed its forecast ES by 14% at 2.5% on P1 (severity ratio 1.14), consistent with normal tails being too thin. The t-GARCH models breach too often (GJR-t 1.39% at 1%), while FHS-GJR, which replaces the t quantile with empirical residual quantiles, breaches 1.13%; one plausible interpretation is that SPY's negatively skewed losses are poorly matched by a symmetric t, but this mechanism was not tested.

Dynamic Quantile test [Exploratory; development only]. DQ asks whether breaches are predictable from past breaches and from the VaR level. Its size was acceptable at 5% (both simulated lengths) and at 2.5% for $T \approx 6{,}500$, but not at 1% (7.8–11.1%) or at 2.5% for short samples, so 1% DQ results are descriptive and P2/P3 at 2.5% are provisional.

Setting (forecasts)	FHS-GJR DQ p, 1%	2.5%	5%	Other ten models
P1 (6,535)	0.019 [descriptive]	**0.002** [Inference]	**0.011** [Inference]	rejected at 2.5% and 5%, all ten
P1 rolling (6,535)	<0.001 [descriptive]	0.073 [Inference]	**0.025** [Inference]	rejected at 2.5% and 5%, all ten
P2 (4,142)	<0.001 [descriptive]	0.041 [provisional]	**0.015** [Inference]	rejected at 2.5% and 5%, all ten
P3 (4,838)	0.14 [descriptive]	0.64 [provisional]	0.43 [Inference]	rejected at 2.5% and 5%, all ten

So at 5%, DQ rejects every model on P1, P2 and the P1 rolling window, and every model except FHS-GJR on P3 [Inference]. FHS-GJR is the least-rejected model (DQ $p < 0.05$ in 8 of 12 cells including the descriptive 1% cells; every other model 11 or 12). For FHS-GJR the predictability comes from the VaR level rather than breach clustering [Descriptive]: on P1 its 1% breach rate is 1.50% when its VaR is below the median and 0.76% when above (overall 1.13%, conditional coverage not rejected): too many breaches when forecast risk is low and too few when it is high.

ES tests [Unsupported test; development only for Z2]. McNeil–Frey and Z2 over-rejected in the calibration study whenever volatility dynamics were misspecified, so their p-values are descriptive. As descriptions: FHS-GJR's P1 severity ratios are 0.989 / 0.975 / 0.981 (realised tail losses close to forecast ES); Z2 is negative for every model on P1 (risk understated on average), smallest in magnitude for FHS-GJR.

5.2 Accuracy (H2)

FZ0 rank among 11 models and mean FZ0 (P1, development) [Descriptive]:

Model	Rank 1%	Rank 2.5%	Rank 5%	Mean FZ0 (1 / 2.5 / 5%)
FHS-GJR	**1**	**1**	2	−3.338 / −3.570 / −3.755
RF-vol + FHS	2	2	**1**	−3.326 / −3.562 / −3.756
GJR-GARCH-t	3	3	3	−3.320 / −3.551 / −3.745
Lasso QR (control)	4	6	10	−3.317 / −3.517 / −3.574

Model	Rank 1%	Rank 2.5%	Rank 5%	Mean FZ0 (1 / 2.5 / 5%)
Quantile forest	9	8	7	−3.057 / −3.422 / −3.648
Gradient boosting	10	9	8	−3.030 / −3.388 / −3.635

- **H2a: no evidence.** The best ML model, the RF-volatility hybrid, has mean FZ0 within 0.012 of FHS-GJR at every level (lower only at 5%, by 0.0015; no economic-significance threshold was pre-specified, so the exact differences are reported), and is worse calibrated (§5.1). The direct-quantile models' mean FZ0 is 0.12–0.31 higher than FHS-GJR's. Draft v1 called these differences significant using Diebold–Mariano; that test was found to over-reject by a factor of 3–10 under realistic persistence (§6.1), so no significance is claimed.
- **H2b: no evidence, descriptively reversed in the deep tail.** At 1% and 2.5% the linear control has lower mean FZ0 than both direct-quantile ML models, and gradient boosting's 1% VaR is breached 2.9× as often as it should be (a rejection by CC, [Inference]). With only about 10–84 training observations below the 1% quantile, trees struggle to learn the extreme tail. ML helped only in the hybrid design, which uses it to model volatility and lets the residuals supply the tail.

5.3 Stress (H3)

The eight crisis episodes cover 1,604 of 6,535 P1 development days; the VIX-above-30 regime covers 688.

Breach rates at 1% by regime (P1) [Descriptive]:

Model	Calm	Crisis episodes	VIX > 30
FHS-GJR	1.05%	**1.37%**	**1.02%**
RF-vol + FHS	1.05%	2.68%	3.05%
Quantile forest	1.03%	3.99%	5.38%
Gradient boosting	2.21%	5.11%	5.52%
HS-250	0.75%	4.05%	4.07%

Loss differential by regime [Descriptive; the HAC regression's p-values are an Unsupported test]. Regressing the FZ0 differential (model minus FHS-GJR) on an episode dummy gives, for the hybrid on P1, a calm-period difference of −0.105 / −0.067 / −0.050 (hybrid better) and a crisis shift of +0.48 / +0.31 / +0.20 (hybrid worse); the two roughly cancel over the full sample, which is why its overall rank ties FHS-GJR. The crisis shift is positive for each of the three ML models in all 36 model × setting × level cells (P1, P2, P3, P1 rolling); these cells are strongly correlated, so this is descriptive consistency, not 36 independent confirmations. A calm-period advantage is specific to the hybrid: its calm difference is negative on P1, P2 and the rolling window and near zero on P3, whereas the quantile forest and gradient boosting have higher loss than FHS-GJR in calm periods too (positive calm difference in 23 of their 24 cells). Draft v1 reported p-values of 0.0015 and 0.015 for these effects; the HAC regression over-rejected at 9–30% under realistic persistence (§6.1), so they are withdrawn.

Reaction speed [Descriptive; 66 largest-loss events on P1]. The day after one of the 1% largest losses, the GJR and FHS-GJR VaR rises 67%; HAR-FHS and EWMA 37–41%; the quantile forest and gradient boosting 23–29% and the RF-vol hybrid 38%. None of the three forest or boosting models reaches 1.5× its pre-shock level within the event window.

H3 status. Relative performance does differ by regime in the descriptive sense: every ML model's loss relative to FHS-GJR worsens in crises in every setting and level, and only the hybrid is ahead in calm periods. No validated test establishes either pattern. Consistency of sign across correlated settings is not a

substitute for a test.

5.4 Interpretability (H4) and the saturation mechanism

Model	C1: top-3 drivers are volatility/VIX	C2: economic signs	C3: stability ≥ 0.6
RF-vol + FHS	✓ (VIX, gk_5, gk_1)	✗ (9/11 positive)	✗ (0.54)
Quantile forest	✓ (gk_5, VIX, pk_1)	n/a (no TreeSHAP for quantiles)	✗ (0.35)
Gradient boosting	✗ (credit-spread change 3rd)	✗ (8/11)	✓ (0.69)

H4 is not met under the pre-stated all-three rule [Descriptive]. The direction is sensible: VIX and range-based volatility dominate, and VIX's contribution rises with its level. But signs of correlated volatility features are unreliable, and importance rankings shift from year to year.

Saturation [Descriptive]. The hybrid in use during 2008 had been trained on data whose maximum VIX was 45.7. Its 1% VaR was 3.81% on 12 September 2008 (VIX 25.7) and 3.82% on 10 October 2008 (VIX 69.9), with identical SHAP contributions: both days fall in the same leaves. FHS-GJR moved from 4.12% to 11.73%. A tree cannot forecast a risk level outside its training range; GARCH extrapolates through its recursion.

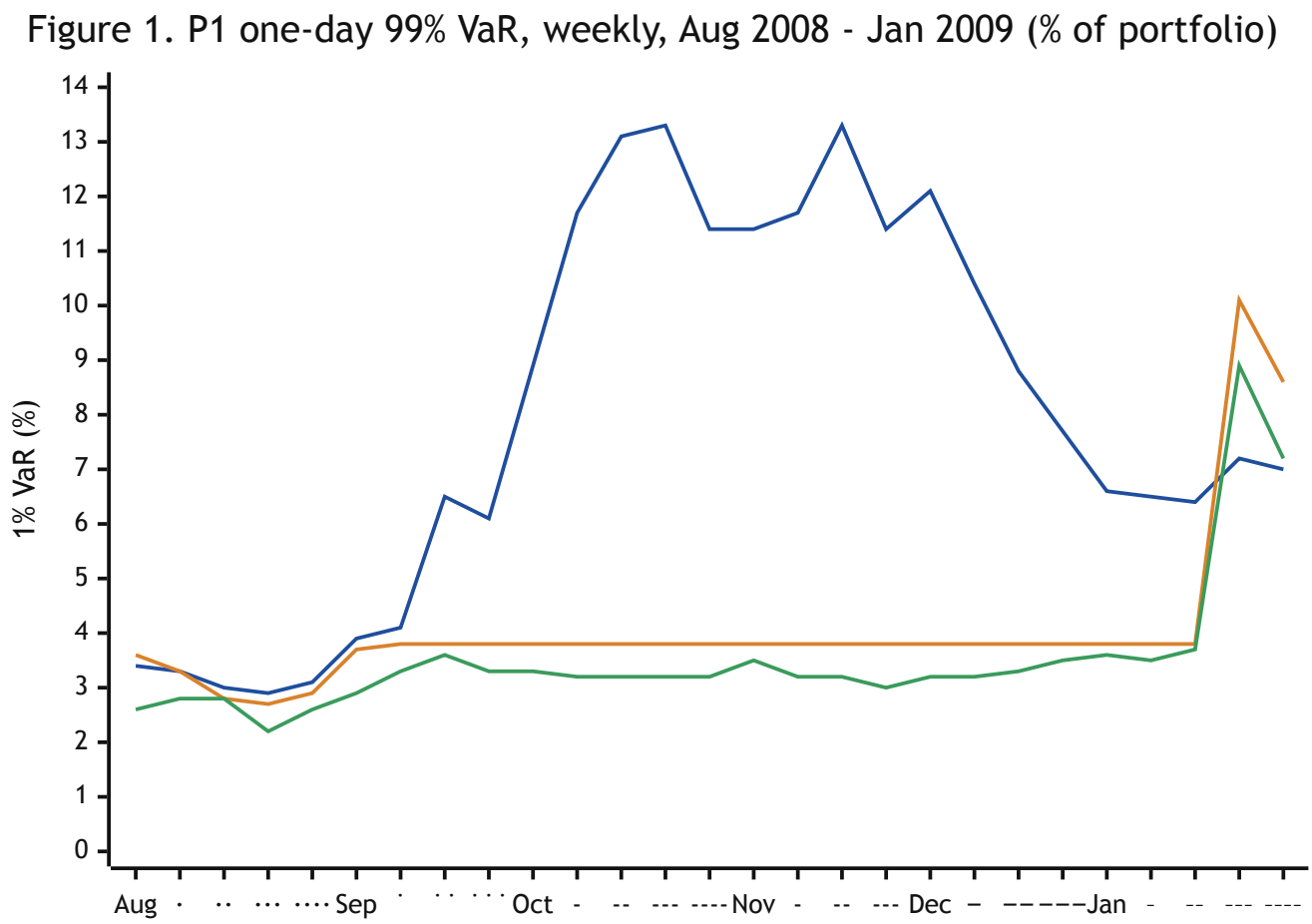


Figure 1. Blue (rising): FHS-GJR. Orange (flat at 3.8%): RF-vol + FHS hybrid. Green (lowest): quantile forest. Ticks are Friday values; month names mark each month's first week. VIX rose from 25.7 (12 Sep) to 79.1 (24 Oct). The forest-based forecasts stay capped until their scheduled yearly refit in January 2009, when the training data first include autumn 2008. Source: `forecasts.parquet` (P1 development run).

5.5 Robustness [Descriptive]

Setting	FHS-GJR FZ0 rank (1/2.5/5%)	RF-vol + FHS rank	Quantile forest / boosting ranks
P1, expanding (main)	1 / 1 / 2	2 / 2 / 1	9, 8, 7 / 10, 9, 8
P2, 60/40	1 / 1 / 1	6 / 7 / 7	10, 9, 8 / 11, 11, 10
P3, Canada	1 / 1 / 1	5 / 3 / 3	8, 9, 9 / 9, 8, 8
P1, rolling 1,000 days	1 / 1 / 1	5 / 5 / 5	10, 10, 8 / 11, 11, 11

FHS-GJR has the lowest mean FZ0 in 11 of 12 cells. The hybrid's calm-period advantage seen on P1 is much weaker elsewhere.

Ten-day horizon [Descriptive; Exploratory, Clarification 8]. Each model's one-day forecast was scaled by √10 (the Basel square-root-of-time rule) and compared with the compounded loss over the next 10 sessions in non-overlapping windows (P1: 653 windows, 6.5 breaches expected at 1%). Scaled FHS-GJR had 7, 16 and 32 breaches against 6.5, 16.3 and 32.7 expected. At 1%, √10 scaling understates risk: breaches exceed the expected count in 41 of 44 model × setting cells (median 1.7×). The counts are too small to test.

5.6 Lockbox 2023–2026: NOT REPLICATED

The lockbox was opened once (protocol commit `dc8af86`, pushed before the run), about 961 forecasts per portfolio. Integrity check: re-forecasting the development period inside the lockbox run reproduced the saved forecasts bit-for-bit for all three portfolios.

The pre-stated rule. The development findings were to be read as *consistent* with the lockbox if (a) FHS-GJR is in the lockbox MCS at 2.5% and 5% **and** (b) neither the quantile forest nor gradient boosting has lower mean FZ0 than FHS-GJR; otherwise *not replicated*.

Result. Condition (b) fails in every portfolio: gradient boosting's mean FZ0 is lower than FHS-GJR's at every level in P1, P2 and P3 (by 0.06 / 0.10 / 0.06, 0.13 / 0.08 / 0.04 and 0.02 / 0.03 / 0.01), and the quantile forest's is lower at 2.5% and 5% in P1 and at all three levels in P2. Because (b) alone is sufficient, the verdict **NOT REPLICATED** holds whatever the MCS does, so it does not depend on the unsupported MCS.

What the verdict does and does not mean. It is a pre-specified descriptive criterion: the development ordering did not carry over to 2023–2026. It does **not** show that ML is statistically better in the lockbox: the differences are small, there are about 10 expected breaches at 1% and 48 at 5%, and the Diebold–Mariano test that would assess them is unsupported (nominal p-values of FHS-GJR against the ML models and the control on P1 ranged from 0.04 to 0.65; descriptive only).

P1 lockbox [Descriptive except CC]:

Model	FZ0 rank (1 / 2.5 / 5%)	Hit rates (1 / 2.5 / 5%)	CC p [Inference]
Lasso QR (control)	1 / 1 / 1	0.73% / 1.56% / 4.37%	0.08 / 0.06 / 0.45
RF-vol + FHS	2 / 3 / 3	1.14% / 2.50% / 4.16%	0.26 / 0.87 / 0.41
Gradient boosting	3 / 2 / 2	1.35% / 2.50% / 5.20%	0.22 / 0.87 / 0.92
Quantile forest	7 / 4 / 4	0.83% / 1.56% / 3.95%	0.13 / 0.06 / 0.15
FHS-GJR	5 / 6 / 5	**1.04% / 1.98% / 4.79%**	**0.89 / 0.37 / 0.63**
GJR-GARCH-t	6 / 7 / 7	1.56% / 2.71% / 6.14%	0.21 / 0.86 / 0.19

Calibration in the lockbox [Inference, with low power]. FHS-GJR's conditional-coverage null is not rejected in any of the 9 portfolio × level cells. The same holds for gradient boosting, the RF-vol hybrid, the lasso control and HAR-FHS (0 of 9 each). With about ten expected breaches at 1%, the lockbox cannot distinguish these models' calibration; it shows only that none of them was detectably miscalibrated in a calm period.

Market conditions [Descriptive; values logged when the lockbox was opened]. VIX was above 30 on 1.6% of lockbox days against 10.5% in development; annualised SPY volatility was 14.9% against 19.9%; there were 4 daily moves larger than 4% against 86. Combined with the stress pattern in §5.3, the reversal is consistent with a regime effect. **That explanation was formed after seeing the result; it is untested and labelled post hoc.** The lockbox contains essentially no stress, so it cannot test stress robustness.

6. Methodological validation

6.1 Finite-sample calibration of the backtests [Exploratory; simulation only]

Why. The ES-regression backtest failed its own validation under dependence (§6.2). That raised the question of whether the other tests are reliable on data like ours. Real P1 FZ0 differentials are extremely heavy-tailed (kurtosis 60 to over 6,000) and persistent (their long-run variance estimated with 100 lags is up to 4.5–5 times their variance for the ML-vs-FHS pairs).

Design (frozen before running; RESEARCH_DESIGN Addition 10). Simulated null worlds in which the hypothesis tested is true by construction: loss differentials with Student-t(3) shocks, stochastic volatility, the real crisis-episode calendar (shock scale doubled, variance quadrupled, in episodes) and a slowly mean-reverting zero-mean component, in designs matched (on autocorrelation and long-run-variance statistics only, never on test outcomes) to the median and upper range of the real series; i.i.d. designs as sanity checks; VaR/ES forecasts that are correct, or correct on average with missed volatility dynamics, or with misspecified tail shape. Sample lengths 6,535 and 961; 2,000 replications; seed 20261112. A test is supported at a given length and level only if its 5% rejection rate is at most 6.4% in every gated design (11.9% for the MCS at 90%). Pre-declared fallbacks: fixed-b Diebold–Mariano (b = 0.1) and a 40-day MCS block. An independent review (Codex) before the run added a harder design and stricter completeness rules; a review after the run confirmed the classifications.

Results (worst gated design; nominal 5% unless stated):

Test	T = 6,535	T = 961	Verdict
Diebold–Mariano (Newey–West)	46.3%	50.1%	Unsupported; fixed-b fallback also fails (7.6% / 20.7%)
HAC stress-dummy regression	30.2%	30.2%	Unsupported
MCS, false exclusion at 90% (limit 11.9%)	98.5%	99.2%	Unsupported; 40-day block also fails
Christoffersen CC	≤ 5.8%	≤ 5.6%	Supported at every level
Christoffersen IND	≤ 6.2%	9.0% at 5%	Supported except T = 961 at 5%
Kupiec POF	≤ 6.4%	7.0% at 1%	Supported except T = 961 at 1%
Dynamic Quantile	7.8% at 1%	11.1% at 1%, 7.5% at 2.5%	Supported at 5% (and 2.5% for T = 6,535)
McNeil–Frey	8.0–14.9% at 2.5–5%	no result in up to 37% of samples	Unsupported in most cells
Acerbi–Székely Z2	13.6–30.7%	13.6–26.5%	Unsupported

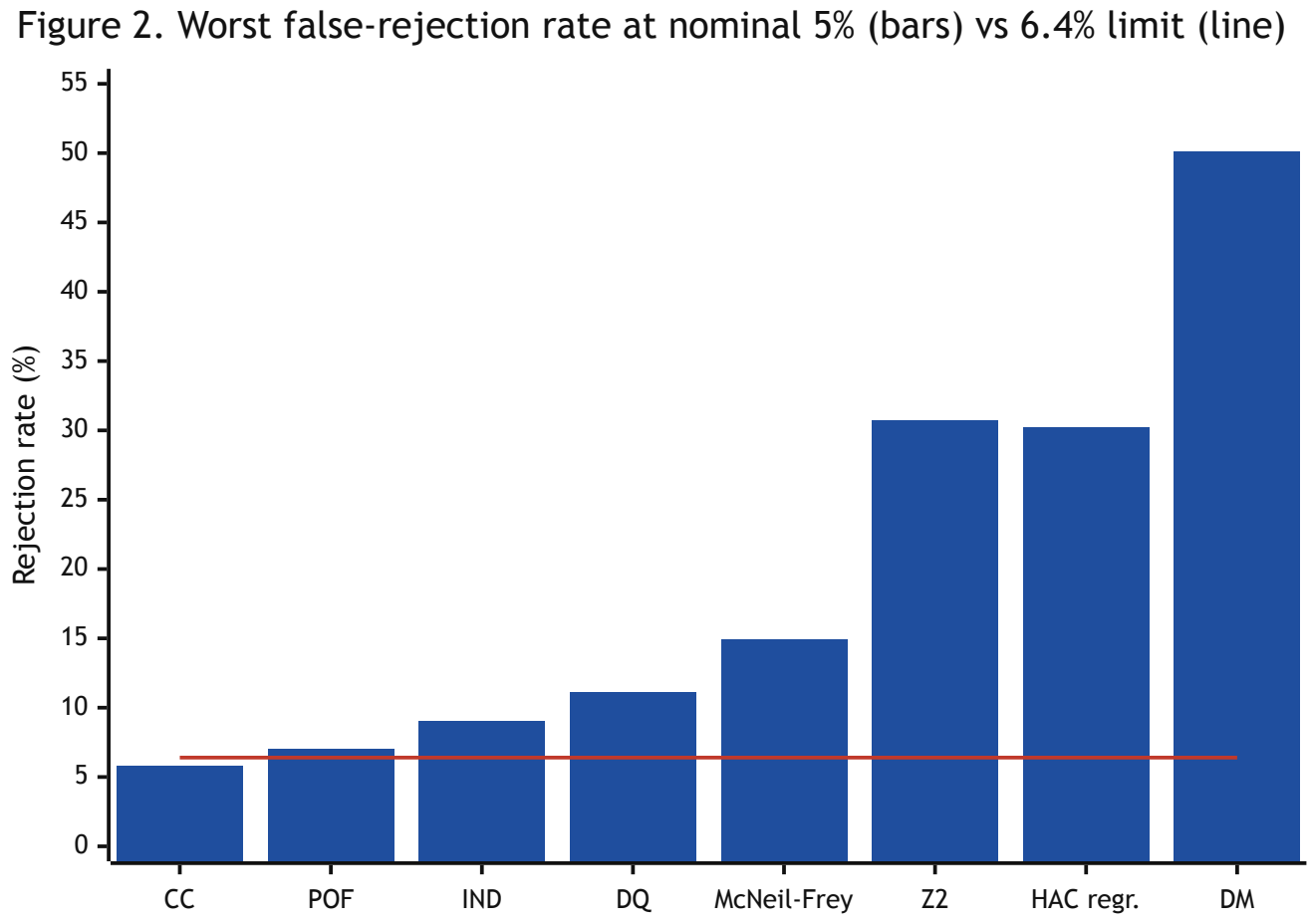


Figure 2. Blue bars: worst rate over designs, sample lengths and levels; red line: the 6.4% limit. The MCS (false exclusion up to 99% at a 10% nominal level) is off this scale.

Interpretation. With independent differentials, DM (4–5%) and the regression (5–6%) are correctly sized, which is consistent with persistence driving the failures; these sanity checks do not by themselves rule out every implementation error. Even the design matched to the *median* real statistics gives DM 14–17%, the regression 9–11% and MCS false exclusions of 39–53%. The magnitude of the distortion depends on persistence beyond lag 100, which a sample of this length cannot pin down; the results show that these tests are vulnerable under designs that match the observed statistics, not the exact distortion in the real series. A "supported" verdict holds only within the simulated designs. Coverage tests are robust for a structural reason: under their null breaches are independent, so dependence is the alternative they are meant to detect, not a nuisance.

6.2 The ES regression backtest: a negative validation result [Exploratory]

The Bayer & Dimitriadis (2022) ES regression (ESR) backtest was added after the lockbox to complement the McNeil–Frey test. I ported the reference implementation (R package `esreg`) to Python and matched it on fixed inputs (to 10 digits once esreg's rule for zero residuals is applied), then checked the bootstrap p-values by simulation before using them:

- The asymptotic p-values over-rejected under serially dependent tail breaches.
- The pre-registered block bootstrap was unusable: its auxiliary scale model ended on a zero-variance boundary in 22–32% of dependent samples, which an audit traced to an unbounded likelihood; 2,479 saved bootstrap draws for one model had been silently degenerate and were quarantined.
- An amended bootstrap (esreg's independence covariance, specified before results) over-rejected at 11.7% and 25.4% (strict test) at nominal 5%, against a 6.4% limit.

ESR p-values were therefore retired; its estimates remain as descriptive diagnostics. Finite-sample size distortions of ES regression tests have already been reported in the literature, so the contribution here is not a new finding about ESR but a documented example of pre-specified validation catching a mis-sized test before it entered the conclusions, and the prompt for §6.1.

7. Discussion

What the evidence supports. Three findings rest on validated tests or on measurements that need no test:

1. The ML models' VaR forecasts fail conditional coverage in the development sample, while FHS-GJR's conditional-coverage null is not rejected in 11 of 12 cells [Inference].
2. At 5%, breaches of every model are predictable (DQ), except FHS-GJR on P3 [Inference]. A well-calibrated average does not mean correct conditional behaviour: FHS-GJR breaches too often when its VaR is low.
3. Tree-based tail forecasts saturate outside their training range (Figure 1), a mechanical property visible in the forecasts themselves.

What the evidence suggests but does not establish. The comparative picture (FHS-GJR's lowest average loss in 11 of 12 development cells, the hybrid's calm-period gain, every ML model's relative loss in crises, the lockbox reversal) is consistent across portfolios and windows, but rests on descriptive comparisons, because the pairwise tests that would assess it are mis-sized on these data. The regime interpretation of the lockbox is post hoc.

What ML is and is not good for in VaR/ES (as a working hypothesis). ML appears more useful as a *volatility* model fed with rich information (VIX, range-based variance) than as a direct *tail* model: there are few tail observations, and trees cannot extrapolate. The hybrid design is the natural architecture to test next.

Practical implications.

- *Bank market risk / model validation:* the results favour retaining a GARCH-filtered historical simulation model as the benchmark and requiring any ML challenger to pass conditional-coverage tests and a stress (out-of-range) check. They also show that model-comparison p-values from standard DM or MCS procedures can be badly over-confident on daily risk data, which matters for challenger-model governance (e.g. OSFI E-23).
- *Pension funds and asset managers* who weight calm-period average accuracy more heavily might use ML-informed volatility in calm regimes, with a guard for stress.
- A regime switch (for example on VIX) or a max(ML, GARCH) rule is the natural next hypothesis. It was **not** tested here, because adding it after seeing the results would be post hoc; it would need new data or a new frozen design.

8. Limitations

1. **Inference on model comparisons.** Diebold–Mariano, the MCS and the stress regression failed their size checks; comparative conclusions are descriptive. Dependence-robust alternatives (larger-bandwidth fixed-b, subsampling) were not validated.
2. **Scope of the calibration study.** "Supported" holds within the simulated designs; real persistence beyond lag 100 is not identified. P2/P3 lengths were not simulated directly.
3. **Statistical power.** About 65 expected 1% breaches in development but only about 10 in the lockbox; crisis cells at 1% are thin. Non-rejections at 1% are weak evidence.

4. **Univariate portfolio approach.** Each portfolio is modelled as one return series with fixed weights; no multivariate (DCC, copula) models.
5. **One market history.** A handful of crises; future crises may differ.
6. **Public data.** Yahoo is unofficial and restates history (immutable snapshots mitigate this); no intraday realized volatility; SPY's daily agreement with the index is weaker before 2009.
7. **Implementation details.** The quantile forest used the library default of one training value per leaf, coarser than Meinshausen's original, and its results depend on the library version (locked). HAR coefficients are attenuated because daily squared returns are noisy proxies. Gradient boosting's per-level quantiles crossed on 66% of days.
8. **Control-model changes.** The lasso control and the safety floor were chosen after seeing the plain linear control fail (Deviation 2); the change strengthens the control, which is conservative for H2b.
9. **Tuning budget.** Small fixed grids, chosen for honesty and compute; richer searches might help ML.
10. **Post-lockbox analyses.** DQ, Z2, ESR, the ten-day table and the calibration study were added after the lockbox was opened and use development data or simulation only.

Full list: `docs/LIMITATIONS.md`.

9. Industry adaptation (Mitacs partnership)

The pipeline is built to take a partner's own data: the provider interface accepts any price or P&L series, portfolios are configured by weights, and every model, test and report is config-driven. A partner project could:

- **Bank (model validation / market risk):** run a challenger study of the partner's 99% VaR and 97.5% ES model with Basel traffic-light and conditional-coverage backtests, and calibrate the comparison tests on the partner's own loss series before relying on them.
- **Pension fund / asset manager:** forecast risk for the actual policy portfolio (P2 is the template), with crisis diagnostics and a pre-registered regime-switching rule tested on new data.

The integrity process (frozen design, committed clarifications, results log, lockbox, pre-specified validation) maps onto model-validation documentation standards.

10. Reproducibility

- **Data:** immutable, hashed snapshots.
- **Code:** a complete git history on GitHub, including the design freeze, the lockbox protocol and every amendment.
- **Environment:** exact package versions in `requirements-lock.txt`. A clean-clone check showed that without it the project was not reproducible one day later (`quantile-forest` 1.5.0 changes forest quantiles).
- **Verification:** a fresh clone with the one-command setup passed the test suite at the time (254 tests) and rebuilt all 11 models' P1 forecasts and backtest tables bit-for-bit. The current suite passes in continuous integration; each fresh-clone check is logged with its exact revision and test count in `RESULTS_LOG.md`.
- **Calibration study:** `python -m alphalab.backtest.calibration {diff|es}` (about 1.3 hours on 4 cores) regenerates §6.1 from its seed.
- **Exploration:** a read-only Streamlit dashboard (`make dashboard`) shows the saved results with evidence-status labels.

11. Conclusion

Under strict out-of-sample discipline, proper scoring rules and pre-committed analysis rules, the ML models studied here did not produce better-calibrated one-day VaR forecasts than filtered historical simulation with GJR-GARCH volatility: their conditional-coverage nulls were rejected almost everywhere in development, while FHS-GJR's was not rejected in 11 of 12 cells. On average loss, no ML model was more than 0.002 better in any P1 development cell, every ML model was relatively worse in crises, only the volatility hybrid was ahead in calm periods, and tree-based tail forecasts saturated outside their training range. A calm held-out period reversed the average-loss ordering, so the pre-stated replication rule returned NOT REPLICATED. Because the standard comparison tests proved unreliable on these data, these comparative findings are descriptive: they are consistent with a regime-dependent role for ML in volatility modelling, but they do not establish the superiority of any model. The methodological lesson is that backtest p-values on daily risk data should be calibrated before they are believed.

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Appendix A: decision timeline (git)

Commit / tag	What it fixed in advance
`d935f1d`	Research design written before any risk result
`prereg-v1` (`46c21ea`)	Design frozen (only HS/EWMA development results seen)

Commit / tag	What it fixed in advance
Deviation 1 (in `100f0a1`)	Rule for failed GARCH refits
Deviation 2 (in `f1b3efc`)	Lasso control and safety floor, after the plain control failed
`2773c4f`	ML specification (Deviation 3), before any ML fit
`a65c036`	Stress rules (Clarification 4), before any stress statistic
`2bb1d58`	H4 criteria (Clarification 5), before any importance value
`b7ab38d`	Robustness specification (Clarification 6), before P2/P3/rolling results
`dc8af86`	Lockbox protocol, pushed to GitHub before the lockbox was opened
`9925df7`	Lockbox results, as observed
`69e8a1c`	Dependency lock after the clean-clone check
`e81e33d`	Z2 specification (Clarification 7), before any Z2 value
`e53085d`	h = 10 specification (Clarification 8), before any 10-day number
`595ed7a`	ESR amendment 10 (bootstrap variant and size gate), before any amended result
`b6c5936`, `b47c95e`	Calibration study design frozen, then revised after independent review, before any run

Appendix B: where each headline number comes from

Claim	Source file (per portfolio run in `data/results/`)
Hit rates, Kupiec/IND/CC p-values	`backtest_coverage.csv`, `backtest_coverage_lockbox.csv`
Mean FZ0, ranks, MCS membership	`backtest_losses.csv`, `backtest_losses_lockbox.csv`
Diebold–Mariano (descriptive)	`backtest_dm.csv`, `backtest_dm_lockbox.csv`
DQ	`backtest_dq.csv`
Severity ratios, McNeil–Frey; Z2	`backtest_es.csv`; `backtest_z2.csv`
Regime breach rates; stress regression	`stress_regimes.csv`; `stress_regression.csv`
Reaction speed	`stress_event_summary.csv`
H4 criteria	`explain_criteria.csv`
Figure 1	`forecasts.parquet`; VIX from the P1 feature build
Ten-day horizon	`horizon10.csv`
Calibration study (§6.1)	`data/results/calibration/{diff,es}_reps2000_seed20261112/`
Lockbox market conditions	`docs/RESULTS_LOG.md`, 2026-11-07 entry